



Presentation

Modernizing Customer Support for a Telco using AI

Farhan Noor Galib

Business Problem: Modernizing Customer Support for a Telco

The Problem

Legacy call center operations create friction at every touchpoint—customers endure **long hold times**, agents struggle with **fragmented information**, and **repetitive issues** drain resources without improving outcomes which ultimately results in **declining CX**

8-12min

High Avg. handle Time

4.2min

Avg. Waiting Time

56%

Ticket Routing Accuracy

65%

First Call Resolution

**No 24/7
Availability**

Constraints

-  **No centralized data repository**
Data stored and managed in silo which was hampering uniformity and authenticity
-  **RBAC Complexity**
Multi-layered access control models requiring department-level, role-based, and document-level permissions.
-  **Integration Complexity**
Must integrate with CRM, Middleware and Ticketing platform and have omni channel experience

Measurable Targets

 Avg. Handle Time **4-5 min**

 Avg. Waiting time **<30 sec**

 Ticket Routing Accuracy **90%**

 Omnichannel Support Availability **24/7**

From Requirements to Technical AI Roadmap

Phased Delivery Approach

1 Discovery

- Workshops with business, agents, IT & compliance
- Data Analysis (Historical tickets, Complain data, Customer conversation etc.)
- KPI Review

2 Pilot

- Develop MVP
- Internal rollout, feedback loop

3 Production

- Scale, optimize, monitor

Outcome

Phased approach de-risked delivery, strategic decisions optimized for enterprise constraints, and risk mitigation ensured cost control and safety.

Strategic Decisions

LLM combined with RAG

Chose LLM for content generation and RAG for accuracy of the content

Data Platform

For centralized data access and future analytics use case

Micro service architecture

API driven service development for better manageability

Centralized knowledgebase

Selected for shared, secured vector search capabilities

Risk Mitigation

Cost Gating

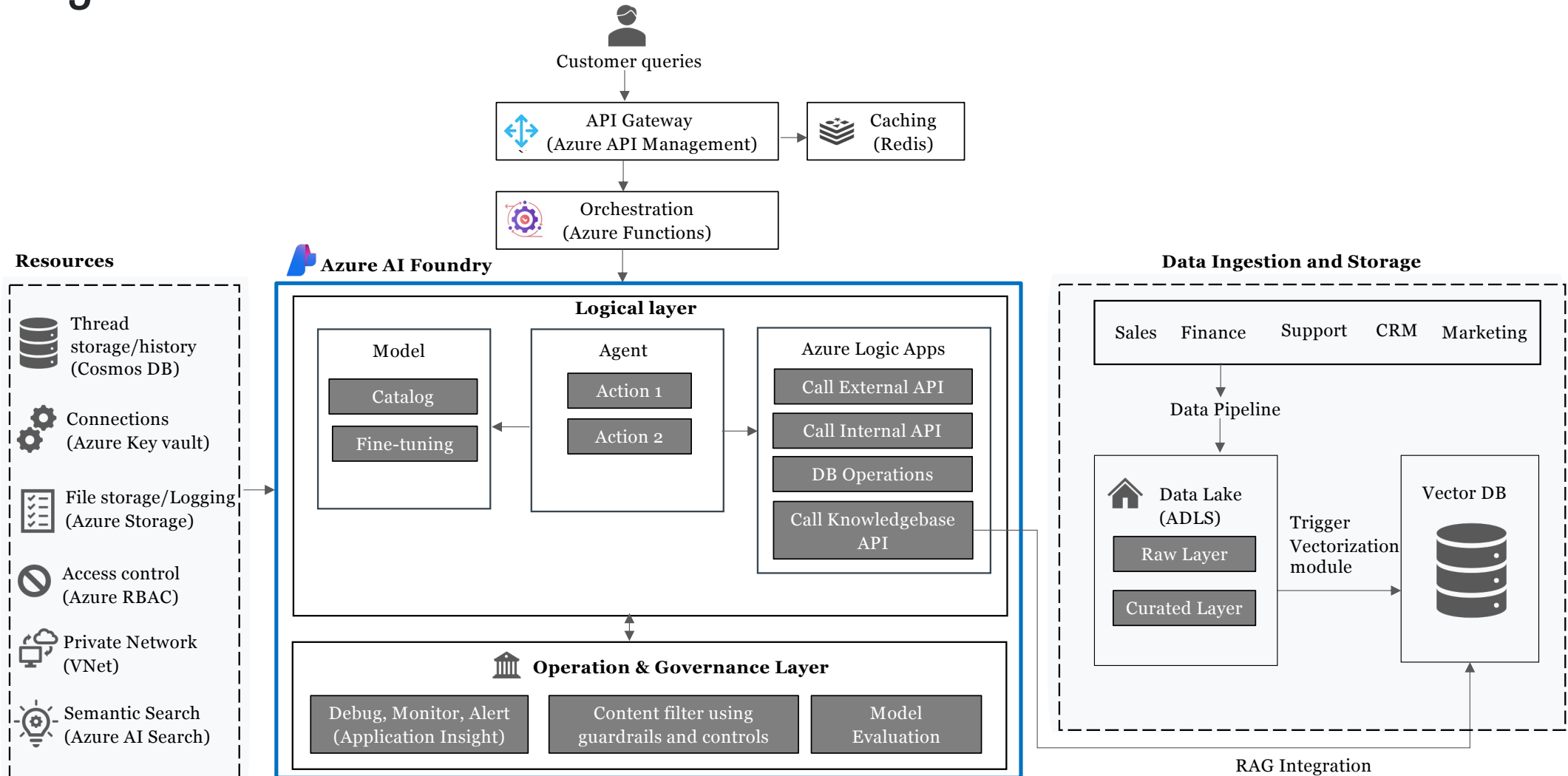
Token limits, query quotas, and budget alerts prevent runaway costs

Content Safety

Data Anonymization and harmful content filtering

Architecture

High-Level Solution Architecture: Conversational AI Platform



Production Deployment: CI/CD, Monitoring & Rollback Strategy



CI/CD & IaC

Trunk-Based Development

Merges to **main** branch trigger a pipeline that validates infrastructure and agent logic

Multi-Stage Pipelines

Azure DevOps pipelines with build, test, security scan, and deploy stages

Infrastructure as Code

All resources (Foundry Project, APIM, AI Search) are defined in Bicep/Terraform

Dev → Test → Prod promotion



Deployment Strategy

Blue/Green Deployment

Parallel environments with instant traffic switch for zero-downtime releases using API Management

Canary Releases

Gradual traffic shifting (5% → 25% → 100%) with automated rollback triggers

Multi region resource distribution

Deploy an identical project in secondary region



Observability

SLO Monitoring

Latency (p95 <2s), availability (99.9%), and accuracy (relevance score >0.85)

Cost Tracking

Per-query cost attribution, budget alerts, and anomaly detection via Azure Cost Management

Alerts

Embedding drift, retrieval efficacy monitoring, and model performance degradation alerts



Rollback & Failover

Blue/Green Agents

Deploy a new agent ID as "Green." Route 10% of traffic via APIM. If metrics are stable, flip the traffic completely

Regional Failover

Deploy a secondary stack in a different region. Since Cosmos DB is globally distributed, agent's memory (Threads) will be available even if the primary region goes down

Uptime

Rollback

Responsible AI, Compliance & Auditability



Safety & Governance Framework



Content Safety

Multi-layered filters for hate speech, violence, self-harm, and sexual content with configurable thresholds



Data Residency

Strict enforcement of data residency within designated zones with no cross-border transfers, meeting regulatory requirements



PII Protection

Automated detection and redaction of personally identifiable information in inputs and outputs



Multi layer authentication

Validating customer identity using multi layer authentication for unauthorized access to sensitive information



Compliance & Auditability



Comprehensive Audit Trails

Every query, response, and decision logged with user context, timestamp, and model version for full traceability



Automated Evaluations

Continuous model evaluation for groundedness, relevance, coherence, and fluency metrics



Human-in-the-Loop

Expert review workflows for sensitive scenarios, low-confidence responses, and compliance-critical queries



Feedback Collection

Built-in thumbs up/down with comment capture, driving continuous model improvement

Reusability & Acceleration: Building the Foundation for Scale

"Build Once, Reuse Everywhere"

Foundational philosophy: every component is designed for reuse across teams and use cases.

Reusable Assets



Standard RAG Engine

Chunking strategy, embedding model, Vector store interface, Retrieval logic



Reusable Service

Agent capabilities are exposed through API call for using future use cases



Infrastructure as Code

Reusable Terraform/Bicep modules for standardized environments



Connectors

CRM connector, Ticketing connector, Knowledgebase connector



Data Pipelines

ETL pipelines for multiple systems



Policy Library

Encapsulate all the policy in a library which can be used in similar AI ecosystem

Thank You